

SELECTING A PUMP FOR CONTINUOUS DUTY

PUMP APPLICATION GUIDE

A decision framework for turning a real process duty into a pump package that can be specified, commissioned, and maintained. Start with the operating profile, not a catalogue point.

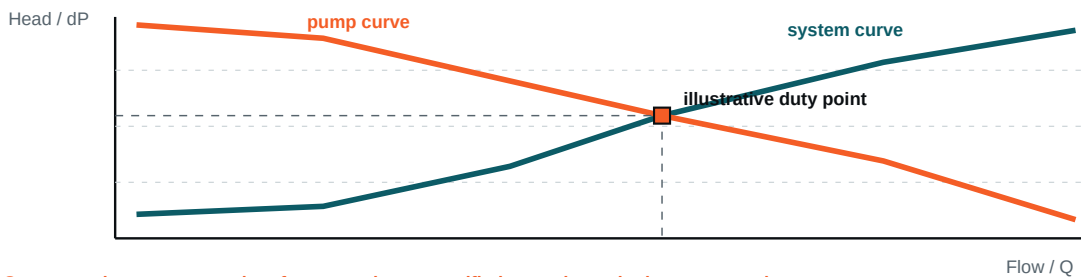
PRIMARY AUDIENCE	REFERENCE STATUS
Process engineers / Package engineers / Operations teams	Illustrative project concept

FIELD SEQUENCE

ENGINEERING LENS	WHAT TO RESOLVE
01 / Define the duty	Record normal, minimum, maximum, and upset flow; differential pressure; fluid temperature; density and viscosity; operating hours; start and stop frequency; and the consequence of losing flow.
02 / Build the system curve	Account for static head, pipework, fittings, control valves, filters, heat exchangers, elevation, and the expected operating range. Keep design assumptions visible so they can be challenged.
03 / Check suction conditions	Review available NPSH across temperature and flow cases. Include upstream losses, tank level, vapor pressure, strainers, and transient conditions rather than relying on one steady-state estimate.
04 / Match materials and sealing	Confirm compatibility with concentration, solids, temperature, corrosivity, and cleaning regime. Select the seal and bearing approach with the maintenance plan in view.
05 / Close the package	Coordinate driver, controls, minimum-flow protection, instruments, isolation, baseplate, access, test basis, spares, and commissioning data as one package.

CONCEPTUAL SYSTEM-CURVE CHECK

The duty point is the conversation between the pump curve and the installed system curve.



Concept only - no measured performance data or certified operating point is represented.

SELECTING A PUMP FOR CONTINUOUS DUTY / HANDOVER RECORD

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A useful field guide ends with evidence that the system is ready for the next team. Capture the records below in the project handover pack and keep exceptions visible until they are closed.

RECOMMENDED RECORDS

01	Duty and upset case table
02	System curve assumptions
03	NPSH and suction review
04	Materials and sealing basis
05	Package and commissioning data sheet

REFERENCE NOTE

This guide supports early planning and field conversations. It does not replace the approved drawings, equipment instructions, site procedures, inspection plan, or project acceptance criteria.